ch25 Radiation and Radioactivity

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2021 Q31

31. A radioactive nuclide plutonium-239 ($^{239}_{94}Pu$) becomes a stable lead-207 isotope ($^{207}_{82}Pb$) after a series of α and β decays. Find the number of β decays in the process.

A. 3
B. 4
C. 5
D. 6

2016 Q32

32. Which of the following statements about ionizing radiations is/are correct ?

(1) The ionizing power of α-particles is much stronger than that of β-particles.
(2) γ-radiation can be completely shielded by a 10 cm thick concrete wall.
(3) Ionizing radiations α, β and γ all undergo deflection in an electric field.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

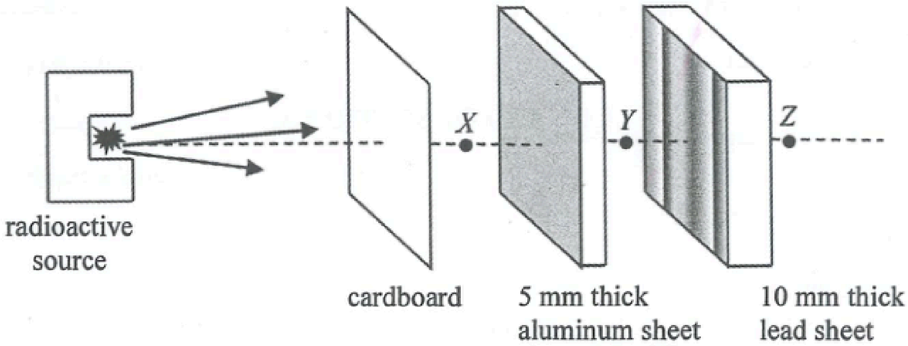
2013 Q34

34. $^{238}_{92}U$ undergoes α-β-β-α decay and becomes a nuclide X. What are the atomic number and mass number of X ?

	atomic number	mass number
A.	90	230
B.	90	234
C.	88	230
D.	88	234

2019 Q31

31. A radioactive source emits α, β and γ radiations.

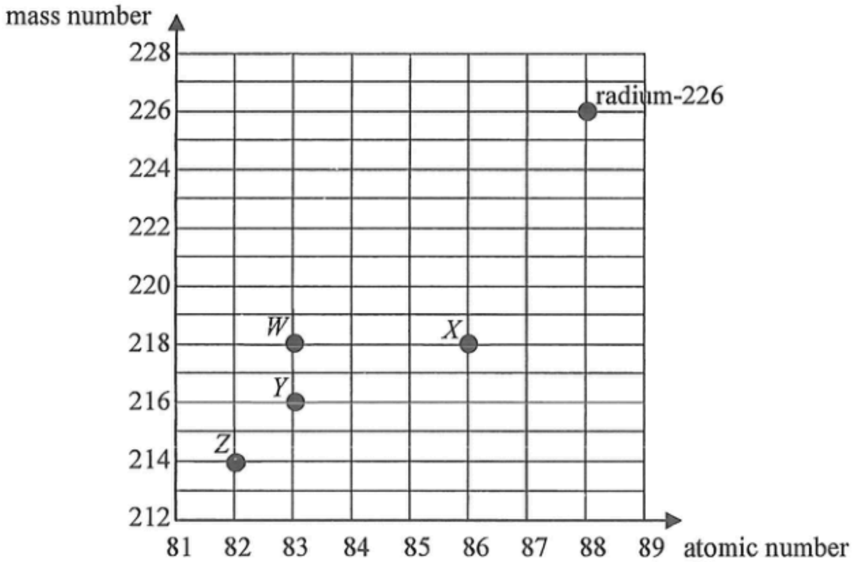


Which statement about the radiation(s) detected at positions X, Y, Z indicated in the figure is correct ?

A. No radiation from the radioactive source is detected at Z.
B. Both β and γ radiations can be detected at Y.
C. α radiation can only be detected at X but not at Y and Z.
D. β radiation can only be detected at X but not at Y and Z.

2025 Q32

32. Radium-226 undergoes a series of α and β decays before finally becomes a stable nuclide. Referring to the plot of mass number against atomic number below, which nuclides, W, X, Y and Z, **cannot** be possible members in the decay series of radium-226 ?



A. W and X
B. W and Y
C. Z and X
D. Z and Y

2017 Q32

32. Which of the following statements about β particles and γ rays is correct ?

A. Only β particles can ionize air particles.
B. Only γ rays can travel through vacuum.
C. Both of them can be detected by a photographic film.
D. Both of them carry charge.

2020 Q30

30. The background count rate in an experiment is determined using a GM counter. Four readings of the count rate in each minute are taken. Which set of readings below is the most probable ?

	1st minute	**2nd minute**	**3rd minute**	**4th minute**
A.	5	62	8	69
B.	40	40	40	40
C.	60	50	30	20
D.	29	26	31	35

2022 Q31

31. Which of the following statements about X-rays is **INCORRECT** ?

A. X-rays can be produced by high-speed electrons hitting a metal target.
B. X-rays are a kind of electromagnetic waves.
C. X-rays can be deflected by an electric field.
D. Although X-rays do not carry charges, they can cause ionization.

2014 Q32

32. A GM counter is placed close to and in front of a radioactive source which emits both α and γ radiations. The count rate recorded is 450 counts per minute while the background count rate is 50 counts per minute. Three different materials are placed in turn between the source and the counter. The following results are obtained.

Material	**Recorded count rate / counts per minute**
(Nil)	450
cardboard	x
1 mm of aluminium	y
2 mm of lead	z

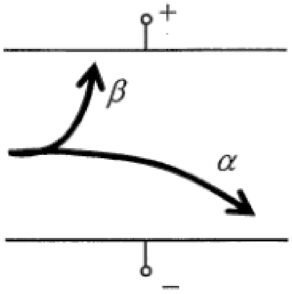
Which of the following is the most suitable set of values for x, y and z ?

	x	y	z
A.	300	300	100
B.	300	100	50
C.	100	100	0
D.	100	50	50

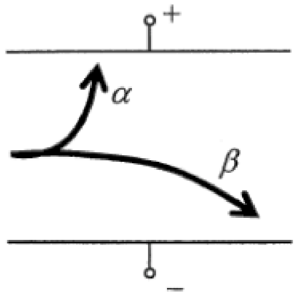
2017 Q31

31. Which of the following diagrams best shows the deflection of α and β particles in a uni field in vacuum ?

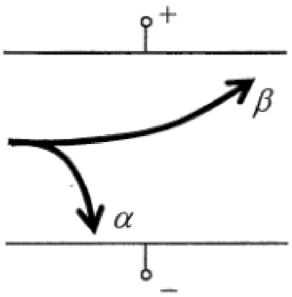
A.



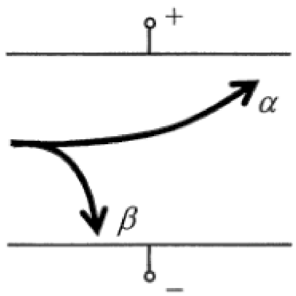
B.



C.

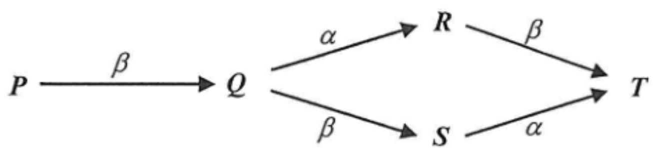


D.



2020 Q31

31.



Nuclide $\boldsymbol{P}$ can decay into nuclide $\boldsymbol{T}$ through either process $\boldsymbol{P} - \boldsymbol{Q} - \boldsymbol{R} - \boldsymbol{T}$ or process $\boldsymbol{P} - \boldsymbol{Q} - \boldsymbol{S} - \boldsymbol{T}$ as shown. Which deductions below are correct ?

(1) $\boldsymbol{P}$ and $\boldsymbol{T}$ are isotopes of the same element.
(2) $\boldsymbol{Q}$ and $\boldsymbol{S}$ have the same number of protons.
(3) $\boldsymbol{S}$ has one more neutron than $\boldsymbol{R}$.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2014 Q31

31. Nucleus W decays to nucleus Z as shown below:

$$W \xrightarrow{\alpha} X \xrightarrow{\beta} Y \xrightarrow{\beta} Z$$

Which of the following statements is/are correct ?

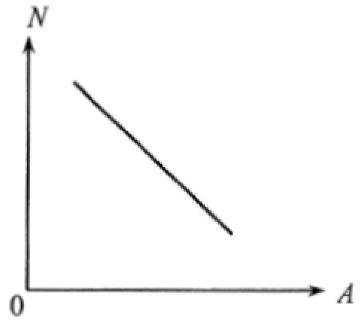
(1) Nucleus X has 1 more proton than nucleus Y.
(2) Nucleus W has 2 more neutrons than nucleus X.
(3) W and Z are isotopes of the same element.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

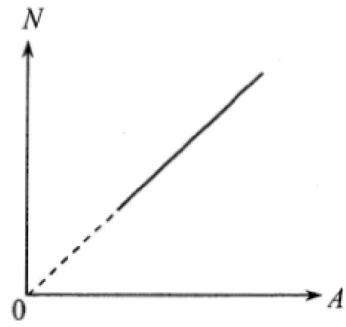
2012 Q36

36. Isotopes of an element have different mass number A and neutron number N. Which of the following N - A plots correctly shows the relationship of N and A for any given element ?

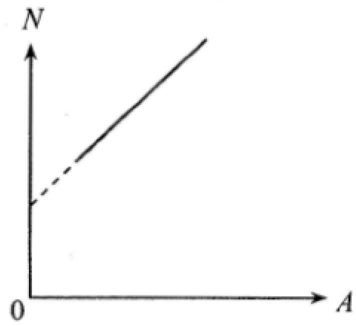
A.



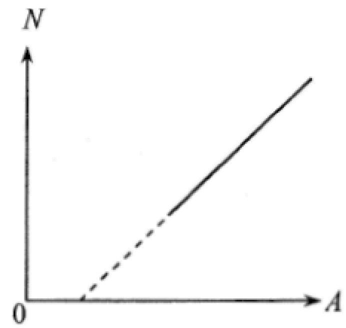
B.



C.



D.



2021 Q33

33. Which of the following may contain sources of ionizing radiations ?

(1) sea water
(2) a rock sample
(3) human body

A. (1) only
B. (2) only
C. (2) and (3) only
D. (1), (2) and (3)

2015 Q31

31. Which of the following nuclear reactions is/are **spontaneous** reaction(s) ?

(1) $^{24}_{11}Na \rightarrow ^{24}_{12}Mg + ^{0}_{-1}e$

(2) $^{10}_{5}B + ^{1}_{0}n \rightarrow ^{7}_{3}Li + ^{4}_{2}He$

(3) $^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + ^{1}_{0}n$

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

pp Q34

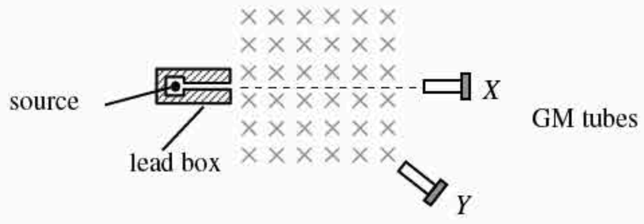
34. Which of the following statements about α and β particles is/are correct ?

(1) The mass of an α particle is greater than that of a β particle.
(2) α particles have a stronger penetrating power than β particles.
(3) An α source can discharge a positively charged metal sphere nearby.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

pp Q35

35.



sap Q36

36. Different absorbers are placed in turn between a radioactive source and a Geiger-Muller tube. Three readings are taken for each absorber. The following data are obtained:

Absorber	**Count rate / s^{-1}**		
–	200	205	198
paper	197	202	206
5 mm aluminium	112	108	111
25 mm lead	60	62	58
50 mm lead	34	36	34

What type(s) of radiation does the source emit ?

A. β only
B. γ only
C. β and γ only
D. α, β and γ